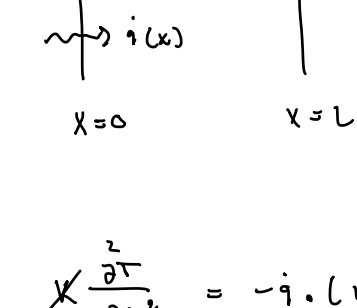


HW 5

Saturday, October 11, 2025 6:53 PM

1) 3.77



$$K \frac{\partial T}{\partial x} \bigg|_{x=0} = -\dot{q} \cdot (1 - \frac{x}{L})$$

$$\frac{\partial T}{\partial x} = \frac{-\dot{q} \cdot x}{K} + \frac{\dot{q} \cdot x^2}{2 \cdot L \cdot K} + C_1$$

$$T(x) = \frac{-\dot{q} \cdot x^2}{2 \cdot K} + \frac{\dot{q} \cdot x^3}{3 \cdot L \cdot K} + C_1 x + C_2$$

$$T(0) = T_0 = C_2$$

$$T(L) = \frac{-\dot{q} \cdot L^2}{2 \cdot K} + \frac{\dot{q} \cdot L^3}{6 \cdot L \cdot K} + C_1 L + T_0$$

or

$$K \frac{\partial T}{\partial x} \bigg|_{x=L} = 0$$

$$\frac{-\dot{q} \cdot L}{K} + \frac{\dot{q} \cdot L^2}{2 \cdot L} + C_1 K = 0$$

$$-\dot{q} (L - \frac{L}{2}) = -\dot{q} \frac{L}{2}$$

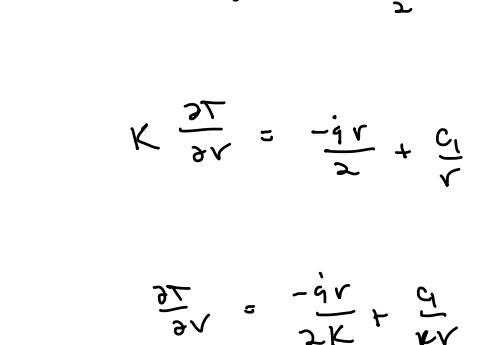
$$C_1 K = 0$$

$$C_1 = \frac{\dot{q} \cdot L}{2K}$$

$$T(x) = \frac{-\dot{q} \cdot x^2}{2 \cdot K} + \frac{\dot{q} \cdot x^3}{6 \cdot L \cdot K} + \frac{\dot{q} \cdot x L}{2K} + T_0$$

$$0 < x < L$$

2) 3.80



$$K \frac{\partial T}{\partial r} \bigg|_{r=r_i} = 0$$

$$K \frac{\partial T}{\partial r} \bigg|_{r=r_o} = h(T(r_o) - T_\infty)$$

• T(r)

• 0 < r < L

$$\frac{1}{r} \frac{\partial}{\partial r} \left(K r \frac{\partial T}{\partial r} \right) = -\dot{q}$$

$$K r \frac{\partial T}{\partial r} = \frac{-\dot{q} r^2}{2} + C_1$$

$$K \frac{\partial T}{\partial r} = \frac{-\dot{q} r}{2} + \frac{C_1}{r}$$

$$\frac{\partial T}{\partial r} = \frac{-\dot{q} r}{2K} + \frac{C_1}{K r}$$

$$T(r) = \frac{-\dot{q} r^2}{4K} + \frac{C_1}{K} \ln(r) + C_2$$

but

$$-\dot{q} r_i + \frac{C_1}{r_i} = 0$$

$$C_1 = \frac{\dot{q} r_i^2}{2}$$

$$C_1 = \frac{\dot{q} r_i^2}{2}$$

but

$$-\dot{q} r_o + \frac{C_1}{r_o} = \frac{-\dot{q} r_o^2}{4K} + \frac{h C_1}{K} \ln(r_o) + C_2 h - h T_\infty$$

$$\frac{h r_o}{K} + \frac{-\dot{q} r_o + \frac{C_1}{r_o}}{h} + \frac{\dot{q} r_o^2}{4K} - \frac{C_1}{K} \ln(r_o) = C_2$$

now plug C₁ into C₂

and plug both back in T(r)

$$\dot{q} \cdot A = \dot{q}'(r)$$

$$\left[\frac{u}{m \cdot s} \right] \left[\frac{m^2}{s} \right] = \left[\frac{u}{m} \right]$$

heat rate per unit length

$$\dot{q} \pi r^2 = \dot{q}'(r)$$

or shell

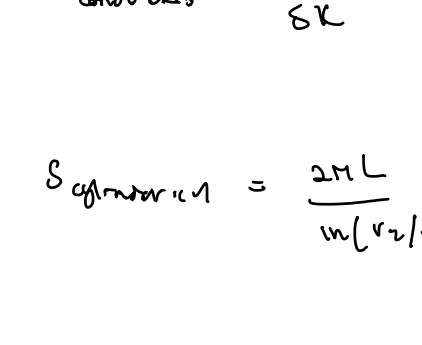
$$\dot{q} \pi (r^2 - r_i^2) = \dot{q}'(r)$$

$$\dot{q} \pi (r_o^2 - r_i^2) = \dot{q}'(r_o)$$

total heat rate

$$Q_r = \dot{q}'(r_o) L$$

3) 4.30



$$T_\infty = 25^\circ C$$

$$h = 50 \frac{W}{m^2 K}$$

$$r = 2.5 mm$$

2D

$$T(r, z)$$

h is

$$T(r, z=0) = T_b$$

$$K \frac{\partial T}{\partial r} \bigg|_{r=2.5} = h(T - T_\infty)$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(K r \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(K \frac{\partial T}{\partial z} \right) + \dot{q} - \dot{q}' = 0$$

1D

$$\dot{q} = SK \Delta T$$

single bar

$$R_{conduction} = \frac{1}{SK}$$

$$S_{symmetry} = \frac{2\pi L}{\ln(r_o/r_i)}$$

$$\dot{q} = -KA_c \frac{\partial T}{\partial x}$$

$$\dot{q} = -KA_c (T_1 - T_2)$$

$$\dot{q} = \sqrt{h P K A_c} \theta$$

$$\theta = T - T_\infty$$

$$T_b = T_b - T_\infty$$

$$m^2 = \frac{hP}{KA_c}$$

$$P = \pi D, A_c = \frac{\pi D^2}{4}$$

$$= 0.017 m, = 0.00002 m$$

R₁ radiation

$$R_f = \frac{\theta}{\dot{q}}$$

$$T(r, z=0) = 100^\circ C$$

$$K \frac{\partial T}{\partial r} \bigg|_z = h(T - T_\infty)$$

$$\theta = T - T_\infty$$

$$m^2 = \frac{hP}{KA_c}$$

$$m = \sqrt{\frac{hP}{KA_c}}$$

$$\theta(x) = T_b - T_\infty$$

$$m = \sqrt{h P K A_c} \theta_b$$

$$q_f = \sqrt{h P K A_c} \theta$$

for A1

$$T_b = \frac{100 + 25}{2} \left| \frac{T_b - T_\infty}{T_b - T_j} \right| = 62.5^\circ C \left| \frac{335.5 K}{335.5 K} \right|$$

for

A1

$$\theta = 37.5^\circ C = 37.5 K$$

$$q_f = 1.23 W \text{ for A1}$$

for S.S

$$m \text{ diff } \theta = \frac{\partial^2 \theta}{\partial x^2} - m^2 \theta = 0$$

$$m \text{ book } \frac{\theta}{\theta_b} = \frac{T_b - T_\infty}{T_b - T_j}$$

$$\frac{\theta_b}{\theta_m} = \frac{\theta_b}{\theta_b} = \frac{1}{J_0}$$

$$\theta_{m1} = \frac{1}{J_0}$$

$$\frac{\theta_b}{\theta_m} = \eta$$

$$\eta = \frac{100 - 25}{T_b - 25^\circ C}$$

$$T_b = 40^\circ C \text{ for S.S}$$

$$q_f = 0.92 W \text{ for S.S}$$

$$R_{con} = \frac{\theta}{\dot{q}} = 16.27 \frac{K}{W}$$

$$R_b = 3 \times 10^{-5} / A_c = 1.6 K/W$$

$$q_b = q_c, R_b \text{ and } R_f \rightarrow (R_f) R_c + R_b = R_T$$

$$R_{A1}: R_b \text{ and } R_c \rightarrow 21.8 K = R_T$$

$$q = \frac{\theta}{R_T} = \frac{100 - 25^\circ C}{21.8 K} = 2.2 W$$

$$q = \frac{T - T_\infty}{R_c}$$

$$T = q R_c + T_\infty = 333.14 K / 60.8^\circ C$$

for A1

for S.S

$$R_f = 16.27 \frac{K}{W}$$

$$R_b = 1.6 K/W$$

$$R_T = \frac{R_f + R_b}{S} = 82.45 \frac{K}{W}$$

$$q = 0.104 W$$

$$T = 312.73 K / 39.78^\circ C$$

$$q = \frac{\theta}{R_T} = \frac{\theta}{R_c + R_b} = \frac{75}{\frac{(R_{f1})}{h A_c} + 1.6}$$

$$\text{for A1 } (R_{f1}) = 2$$

$$\text{for S.S } (R_{f1}) = 5$$

graph:

Al vs SS with h

